

Answer key

Multiple choices

1. B
2. C
3. C
4. B
5. D
6. G
7. F
8. B
9. A
10. D
11. C
12. A
13. D
14. A
15. E

Short answers:

1. The three examples of aqueous solutions are, solid solute: sugar in water liquid solute: alcohol in water and gas solute: carbon dioxide in water.
2. A volumetric flask and a volumetric pipette, or graduated pipette, can be used to prepare a solution with a known concentration.
3. A can of pop at room temperature is more likely to fizz and spray at opening than a can from refrigerator. Water at higher temperature can keep less dissolved gasses and they go out, because solubility of gasses decreases with temperature.
4. Waste water usually is at higher temperature than ambient water, furthermore organic compound in waste water undergo bacterial fermentation, which is exothermic process. High temperature decreases the concentration of oxygen in the lake, because solubility of gasses decreases with temperature. Low oxygen concentration causes respiratory problems for fish.
5. (a) Diluting acidic solutions decreases the hydrogen ion concentration. This increases the pH and makes these solutions less acidic.
(b) Diluting basic solutions decreases the pH and makes these solutions less basic.
6. *Acids*
 - tastes sour
 - turns blue litmus red
 - reacts with active metals to produce hydrogen gas*Bases*
 - tastes bitter
 - feels slippery
 - has a pH higher than 7
 - neutralizes acids

Essay answers:

1. Cooks, bakers, laboratory technicians, chemical engineers and others who work with solutions apply a variety of techniques and instruments to increase the rate of dissolving.
The rate of dissolving is a measure of how quickly a solute dissolves in a solvent. The factors that affect the rate of dissolving are agitation/mixing, temperature and surface area.
Grinding increase the surface area and by that the rate of dissolving.

The higher is surface area of a solute, the higher is the amount of solute that is in direct contact with the solvent, and thereby the frequency of collisions with the solvent molecules is bigger, making the solute to dissolve more rapidly.

For grinding solids may be used mortar and pestle, grinder and other techniques.

2. Following issues should be addressed in the essay:

- Process was designed for production of *baking soda* NaHCO_3 and *washing soda* Na_2CO_3
- The major discovery was
 - Reaction

$$\text{NH}_4\text{HCO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{NH}_4\text{Cl}(\text{aq}) + \text{NaHCO}_3(\text{s})$$
 - Solubility of NaHCO_3 (sodium hydrogen carbonate) is considerably lower than NH_4Cl (ammonium chloride)
 - Difference in solubility was used to separate baking soda by crystallization.

Problem based questions:

1. Measure 14 g of the solute and dissolve in water, add water to bring the total volume of the solution to 400mL.
2. $v/v = \text{volume of solute} / \text{volume of solution} \times 100\%$
1.5L
3. 0.02 mol/L
4.

$$2 \text{AgNO}_3 + \text{Cu} \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{Ag}$$
 0.2725 g of Ag^+ will remain in solution
 To precipitate it all Ag^+ we need 0.48 g of Cu or 0.08 g more than it is given
5. Amount of precipitate formed in given reaction is **1.36 g**.
6. **Analysis:** Use solubility chart (Nelson Chemistry 11, 2002, p.324) to determine which anions selectively precipitate only one of these cations.

Solution: Cl^- makes low solubility salt (precipitate) with Cu^+ and SO_4^{2-} with Ca^{2+} , so we can use soluble salt such as NaCl and Na_2SO_4 for qualitative analysis.

